

WHAT THIS MANUAL DOES

This manual walks you from a Windows laptop with nothing installed to a working test rig that can command the steering rack of a 2013 Tesla Model S over CAN bus. Read every step in order. Do not skip ahead.

WHAT YOU NEED

- Windows 10 or 11 laptop, internet connection
- SYS TEC sysWORXX USB-CANmodul1 (model 3204001) -- the black box you have
- A USB-A cable (the SYS TEC has a USB-B port on top)
- Digital multimeter (for the pinout verification procedure)
- CAN H and CAN L wires from SYS TEC DB9 to **verified** Tesla chassis CAN pins
- The 2013 Tesla Model S, key in, ignition on (Drive Ready not required)

STEP 1 INSTALL PYTHON

Python is the language this program is written in. It does not come on Windows by default.

1.1 Open your web browser and go to:

<https://www.python.org/downloads/>

1.2 Click the big yellow button that says **Download Python 3.12.x** (or whatever version is shown). The .exe file will save to your Downloads folder.

1.3 Double-click the downloaded file to start the installer.

1.4 CRITICAL: Before clicking Install, check the box at the bottom that says "Add python.exe to PATH". If you skip this, nothing else in this manual will work.

1.5 Click **Install Now**. Wait for it to finish (about 2 minutes). Click Close.

1.6 Press the **Windows key**, type **cmd**, and hit Enter. A black command prompt window opens.

1.7 Type this exactly and press Enter:

```
python --version
```

You should see something like "Python 3.12.x". If you see "not recognized", you forgot to check the PATH box. Reinstall and check it.

STEP 2 INSTALL PYTHON-CAN

python-can is the library that talks to your SYS TEC adapter. Still in the same black command prompt window, type:

```
pip install python-can
```

Wait for it to finish (10 to 30 seconds). The last line should say **Successfully installed python-can-x.x.x**. Leave the window open for now.

STEP 3 INSTALL THE SYS TEC DRIVER

The SYS TEC box needs a Windows driver. Without it, Python cannot reach the device.

3.1 Go to: <https://www.systec-electronic.com/en/services-support/downloads>

3.2 In the search box, type **USB-CANmodul**. Find the entry called **"sysWORXX USB-CAN Driver for Windows"**. Download the latest x64 version.

3.3 Run the installer as Administrator (right-click, **Run as administrator**). Click through with defaults. Reboot if it asks.

3.4 Plug your SYS TEC box into the laptop with the USB cable. Windows will install the device. Open Device Manager (Windows key + X, then choose Device Manager). You should see a category **USB-CAN Devices** with your unit listed. If it has a yellow warning triangle, the driver did not install correctly -- redo step 3.3.

STEP 4 VERIFY THE PINOUT (SEPARATE PROCEDURE)

STOP. Before you wire anything to the rack or to the SYS TEC, you must verify which pins on your specific 2013 Model S carry chassis CAN. Pin numbering varies across build dates and connector revisions. Connecting CAN H/L to a 12V pin will destroy the SYS TEC adapter.

Open the file **PINOUT_VERIFICATION.pdf** in this folder and complete the entire procedure (Tests A, B, C). It uses a multimeter and the included **can_sniffer.py** program to prove you are on the real chassis CAN bus before any active testing. Do NOT continue past this step until the verification document tells you to.

Wiring summary (after verification): SYS TEC DB9 pin 7 = CAN H, pin 2 = CAN L, pin 3 = GND. Twist H and L together along the run. Bus should read ~60 ohms between H and L with car off.

STEP 5 PLACE THE PROGRAM FILE AND RUN

5.1 The program is the file named **tesla_steering_test.py** in the same folder as this manual.

5.2 Open a command prompt in that folder. Easiest way: open File Explorer, navigate to the folder, click in the address bar, type **cmd**, press Enter. A black window opens, already in the right folder.

5.3 Type:

```
python tesla_steering_test.py
```

5.4 The control window opens. Leave the black command window open behind it -- error messages will appear there if something goes wrong.

STEP 6 FIRST POWER-UP (PASSIVE LISTEN ONLY)

Front of the car must be jacked up with wheels off the ground OR tie rods disconnected from the rack ends. Do not skip this. Even a passive test can wake the rack if power is healthy.

6.1 Turn the Tesla on (key fob in, brake pressed for accessory power; full Drive Ready not needed).

6.2 In the program, click **CONNECT**. Within 2 seconds the top-right indicator should say **EPAS LINK OK** in green and the "0x370 RX" counter should start climbing.

6.3 Have a helper turn the steering wheel by hand. The "Measured Angle" readout should change. If it does, your wiring and bus are good.

6.4 If "0x370 RX" stays at 0 after 5 seconds: wrong pins, wrong baud rate, or swapped CAN H/L. Disconnect, double-check wiring, reconnect.

STEP 7 FIRST ACTIVE COMMAND

Only do this once Step 6 is fully passing. Keep one hand near the laptop ready to hit ESC.

7.1 Click **ENGAGE**. Watch the "EAC Status" field. ACTIVE (green) means the rack will follow commands. FAULT (red) means it refused; read the "Last Error" field.

7.2 If you see **MIN_SPEED**: open **tesla_steering_test.py** in Notepad, change the line **BENCH_MODE = False** to **BENCH_MODE = True**, save, restart the program.

7.3 Move the slider to **+5 deg**. Confirm the rack moves a small amount in the expected direction. Note the sign convention. Do not jump straight to large angles.

7.4 Try +/- 30. Then 0. Then +/- 60. Stop if anything feels wrong.

7.5 To stop at any time: press ESC, click the red E-STOP button, or close the window. The rack will disengage and the steering wheel will go limp.

This rack is already firmware-patched (Derek confirmed). That means the rack ignores the GTW/EPB gating signals and accepts 0x488 directly with valid checksum and counter. The stock GTW and EPB modules in the car are still sending 0x101 and 0x214 -- leave them alone, the rack just needs them present on the bus, contents don't matter after the patch.

TROUBLESHOOTING QUICK REFERENCE

Symptom	Likely cause / fix
"python is not recognized"	PATH not set during install. Reinstall Python with the PATH checkbox ticked.
pip install fails	No internet, or you are behind a corporate proxy. Try a different network.
CONNECT shows red error in the log	SYS TEC driver not installed, or device unplugged. Recheck Step 3.
"EPAS LINK OK" never appears	Wiring problem. CAN H/L swapped, wrong pins, or wrong baud rate.
EAC Status: FAULT, MIN_SPEED	Rack wants vehicle speed. Set BENCH_MODE = True (Step 7.2).
EAC Status: FAULT, HANDS_ON	Someone is touching the wheel. Let go and re-engage.
EAC reaches ACTIVE but rack does not move	Stock GTW or EPB may not be on the bus, or your CAN H/L are flipped. Call Derek.
E-STOP fires repeatedly	Bus errors or RX timeout. Check for a loose wire; the divergence trip is OFF by default.